Assignment 5 - Product Analytics -Uber Driver Dashboard

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Problem Statement

Drivers make dozens of micro-decisions during a shift:

- When to go online

- Where to position themselves
- Which trips to accept
- When to move zones

Today, most drivers rely on intuition, prior experience, or informal community advice. While Uber has rich demand and marketplace data, drivers have limited visibility into the factors that directly influence their earning potential.

As a result, drivers struggle to understand:

- Why some shifts perform better than others
- Where demand is building
- How their behavior impacts earnings
- How they compare with similar drivers

This leads to missed earning opportunities, inconsistent income, and reduced confidence in decision-making.

Driver Pain Points

1. Slow Shift: Bad Luck or Poor Zone/Time Selection?
   Drivers often experience low-earning shifts but have no way of understanding the root cause
2. Limited Visibility Into Future Demand
   Drivers can react to demand but struggle to anticipate it.
3. High Cancellation Rate and Declining Ratings Go Unnoticed
   Many drivers focus on earnings but lack visibility into operational metrics influencing future trip opportunities.
4. No Meaningful Benchmarking
   Most drivers do not know whether they are performing well relative to peers.
5. Income Is Unpredictable
   Drivers often experience significant day-to-day income fluctuations.

Goal

Primary Goal

- Enable drivers to maximize **Earnings per Online Hour** by providing actionable, real-time insights on demand, zone selection, and driving behavior.

Secondary Goal

- Reduce time spent waiting for trips
- Identify high-opportunity zones before demand peaks
- Learn from previous shifts
- Make data-driven decisions instead of relying solely on intuition
- Maintain sustainable driving habits without increasing working hours

Why Earnings per Online Hour?

Drivers ultimately care about earning more from the time they are already willing to spend online. Total earnings alone is an incomplete measure because it can be increased simply by working longer hours. Earnings per Online Hour better reflects efficiency, decision quality, and utilization of earning opportunities.

Wellbeing Constraint

Driver wellbeing is a first-class product requirement.

Any recommendation engine must balance earning opportunities with fatigue indicators such as:

- Consecutive driving hours
- Break frequency
- Weekly online hours

Success should be measured by higher earnings efficiency rather than increased driving duration.

Driver Personas & Benchmarking Strategy

Not all drivers operate the same way. Comparing every driver against the city average can produce misleading insights because driving patterns, availability, and goals differ significantly. To ensure recommendations and benchmarks are relevant, drivers should be compared against peers with similar operating behavior.

Persona	**Full-Time Driver**	**Peak-Hour Driver**	**Weekend Driver**	**Airport Specialist**

Characteristics	• Online 35-60 hours/week • Drives throughout the day • Depends on Uber as primary income source	• Primarily drives during morning and evening rush hours • Optimizes for high-demand periods • Limited daily driving hours	• Drives mostly on weekends • Supplements another source of income • Flexible availability	
Success Metric	• Earnings per online hour • Weekly earnings stability	• Earnings per hour during peak windows • Demand capture rate	• Earnings per active shift • Weekend earnings per hour	• Earnings per trip • Airport trip utilization

Demand Opportunity Score is a predictive metric that combines demand volume, surge probability, waiting-time reduction, and historical earnings into a single 0-100 score representing the earning potential of a zone. The score helps drivers make faster, data-driven decisions about where to position themselves, improving earnings per online hour while reducing idle time.

Solution

Core Principle

A driver's earnings are influenced by two broad factors:

1. **Time spent online**
2. **Quality of decisions made while online**

Increasing online hours can increase total earnings, but it also introduces the risk of fatigue, burnout, and reduced driver wellbeing. Therefore, the dashboard should focus on improving **earnings efficiency** rather than encouraging longer working hours.

Key Assumption

Online hours are treated as relatively fixed.

The objective is not to increase the number of hours a driver works, but to help the driver earn more from the hours they have already chosen to work.

Levers That Influence Earnings per Hour

Based on the available data, the primary variables a driver can control are:

1. Zone Selection

Different areas of a city exhibit different demand patterns, trip density, rider availability, and surge frequency.

Selecting the right zone can result in:

- More trip requests
- Reduced idle time
- Higher surge opportunities
- Higher earnings per hour

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2. Time Selection

Demand fluctuates significantly throughout the day.

Examples:

- Morning commute
- Office closing hours
- Airport arrival peaks
- Weekend nightlife

Choosing to drive during high-demand windows can substantially improve trip availability and earning potential.

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3. Zone + Time Combination

The highest earning opportunities often emerge from a combination of **being in the right location at the right time**.

For example:

Zone	Time	Opportunity
Airport	2 PM	Medium
Airport	8 PM	High
Business District	8 AM	High
Business District	3 PM	Low

Therefore, neither zone nor time alone is sufficient. The dashboard must evaluate them together.

Product Hypothesis

If drivers are provided with timely, personalized recommendations on:

- Where demand is building
- Which zones have the highest earning potential
- When demand is expected to increase
- How their current performance compares to similar drivers

then drivers will make better operational decisions, resulting in:

- Higher Earnings per Online Hour
- Lower Idle Time
- More Consistent Earnings

- Greater Confidence in Decision-Making

Dashboard Recommendation Engine

The dashboard should answer three key questions:

1. How am I doing?

Plain Text

Earnings/hr: ₹210

City Benchmark: ₹190

You are performing 11% above similar drivers

Show more lines

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2. Why am I performing this way?

Plain Text

Strong performance driven by:

✓ High-demand zone

✓ Evening peak hours

✓ Low cancellation rate

Areas for improvement:

⚠ Missed nearby surge opportunity

Show more lines

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3. What should I do next?

Plain Text

Best Opportunity Nearby

Zone: Indiranagar

Demand Opportunity Score: 92

Estimated Earnings:

₹280-₹320/hr

Demand expected to peak in 20 minutes

Demand Opportunity Score

Component	Weight
Demand Volume	40%
Surge Probability	30%
Waiting Time Reduction	20%
Historical Earnings	10%

Demand Opportunity Score =40% Demand Volume + 30% Surge Probability+ 20% Waiting Time Reduction+ 10% Historical Earnings

Metrics

1.North Star Metric - Earnings per Online Hour

Metric	Derived From	Why It Matters
Earnings per Online Hour	**Total Earnings ÷ Total Online Hours**	Measures how efficiently a driver converts working time into income. Unlike Total Earnings, it cannot be improved simply by working longer hours. It captures the impact of location choice, timing, demand, trip acceptance, and

		operational efficiency while supporting sustainable earning behavior.

Reason for being North Star Metric - **Earnings per Online Hour**

Aspect	**Rationale**
Driver Success	Reflects how much income a driver earns for each hour invested.
Actionable	Can be improved by choosing better zones and times, not just by working more hours.
Fair Comparison	Enables comparison across drivers with different working hours.
Aligns with Driver Goal	Answers the driver's primary question: "How do I earn more from the hours I'm already working?"
Supports Wellbeing	Encourages efficiency rather than excessive online hours.
Supports Business Goal	Higher earnings efficiency can improve driver satisfaction and retention.

2. Driver Metrics (Leading Indicators)

Metric	**Derived From**	**Why It Matters**
Demand Score	Combination of ride request volume, active rider demand, surge probability, and expected wait time in a zone/time window	Predicts future earning opportunities. Higher demand typically results in more trip requests, less idle time, and higher earnings per online hour.
Zone Quality	Historical earnings/hour, trip density, demand trends, and surge frequency for a specific zone	Helps drivers identify locations with the highest earning potential before demand peaks, enabling proactive positioning.
Acceptance Rate	(Trips Accepted ÷ Trip Requests Received) × 100	Reflects how consistently a driver accepts available opportunities. A healthy acceptance rate helps maximize trip opportunities and earnings potential.

Cancellation Rate	(Trips Cancelled ÷ Trips Accepted) × 100	High cancellation rates reduce completed trips and may negatively impact matching efficiency, resulting in lost earnings opportunities.
Rating	Average rider rating over a defined period (e.g., 7-day or 30-day average)	Indicates service quality and rider satisfaction. Maintaining a strong rating promotes trust and long-term platform success.
Consecutive Driving Hours	Continuous online/driving time without a significant break	Serves as an early warning signal for fatigue. Long uninterrupted driving sessions can reduce alertness, safety, and overall driving quality.
Break Hours / Break Frequency	Total break duration and number of breaks taken during a shift	Helps ensure adequate recovery time, supports driver wellbeing, and reduces fatigue-related risks.

3. Outcome Metrics

Metric	Derived From	Why It Matters
Earnings Today	Sum of earnings from all completed trips during the current day/shift	Provides a simple snapshot of current shift performance and progress toward the driver's daily income expectations. While important, it should be viewed alongside Earnings per Online Hour to avoid encouraging excessive working hours.
Trips Completed	Total number of successfully completed trips during the selected period	Measures how effectively a driver converts ride opportunities into completed, revenue-generating trips. Helps assess trip volume and operational productivity.
Earnings vs Benchmark	Driver's Earnings per Online Hour compared against similar drivers (same city, vehicle	Provides performance context by showing whether the driver is earning above, below, or in line with comparable drivers. Helps identify opportunities for improvement and

	category, and driving pattern)	validates whether changes in driving behavior are delivering results.

4. Success Metrics

Metric	Derived From	Why It Matters
Increase Earnings/Hour (%)	% change in Average Earnings per Online Hour before vs. after dashboard adoption	Directly measures whether the dashboard is helping drivers earn more from the same amount of working time. This is the primary indicator of product success and alignment with the North Star metric.
Increase Trip Acceptance (%)	% change in Acceptance Rate before vs. after using dashboard recommendations	Indicates whether drivers are acting on demand and zone recommendations, leading to better utilization of available earning opportunities.
Reduce Idle Time (%)	% reduction in time spent online without an active trip	Lower idle time means drivers are spending more time on revenue-generating trips and less time waiting, improving operational efficiency and earnings potential.
Reduce Driver Earnings Variance	Reduction in fluctuations of Earnings per Hour across days, weeks, or shifts	Consistent earnings are often more valuable to drivers than occasional high-income days. Lower variance improves income predictability and driver confidence in planning work schedules.
Improve Driver Satisfaction	Driver survey scores, app feedback, NPS/CSAT, or satisfaction ratings	Measures whether drivers perceive the dashboard as valuable, trustworthy, and helpful in achieving their earning goals. Higher satisfaction can improve driver retention and engagement.

5.Operational Efficiency Metrics

Metric	Derived From	Why It Matters
Deadhead Distance	Distance traveled without passenger	Proxy for fuel consumption and inefficient positioning.

6. Guardrail Metrics

Metric	Derived From	Why It Matters
Fatigue Score	Composite score derived from Consecutive Driving Hours, Break Hours/Frequency, Weekly Online Hours, and (optionally) Late-Night Driving Patterns	Ensures the dashboard does not maximize earnings at the expense of driver wellbeing. Acts as an early warning indicator for fatigue risk and helps promote sustainable driving behavior.
Consecutive Hours Driving	Continuous driving/online time without a meaningful break	Long uninterrupted driving sessions can reduce alertness, increase fatigue risk, and impact driving performance. Used to trigger break recommendations and wellbeing alerts.

Design Consideration

The recommendation engine should avoid continuous re-ranking whenever a driver enters a new zone.

A minimum movement threshold or recommendation refresh interval should be introduced to prevent:

- Driver confusion
- Excessive repositioning
- Unnecessary fuel consumption
- Recommendation loops

Real-Time Dashboard Experience

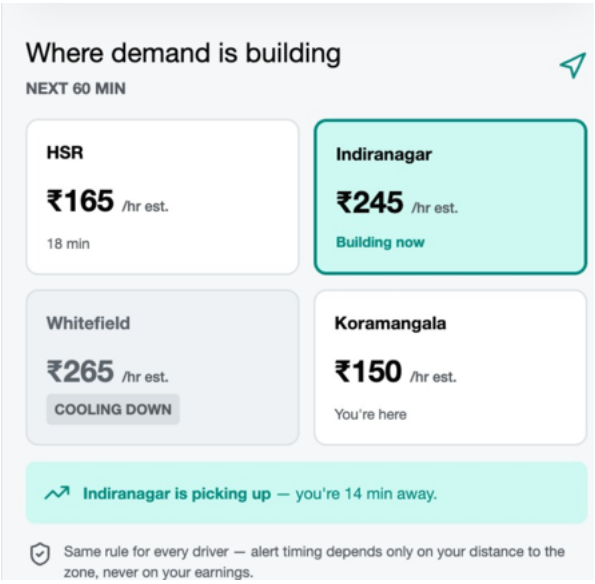
1.Demand Opportunity Map & Nearby Zone Recommendations

The dashboard presents current and nearby zones ranked by earning potential.

The recommendation engine considers:

- Historical earnings
- Current demand
- Surge probability
- Acceptance trends
- Cancellation trends
- Trip density

The driver is shown:



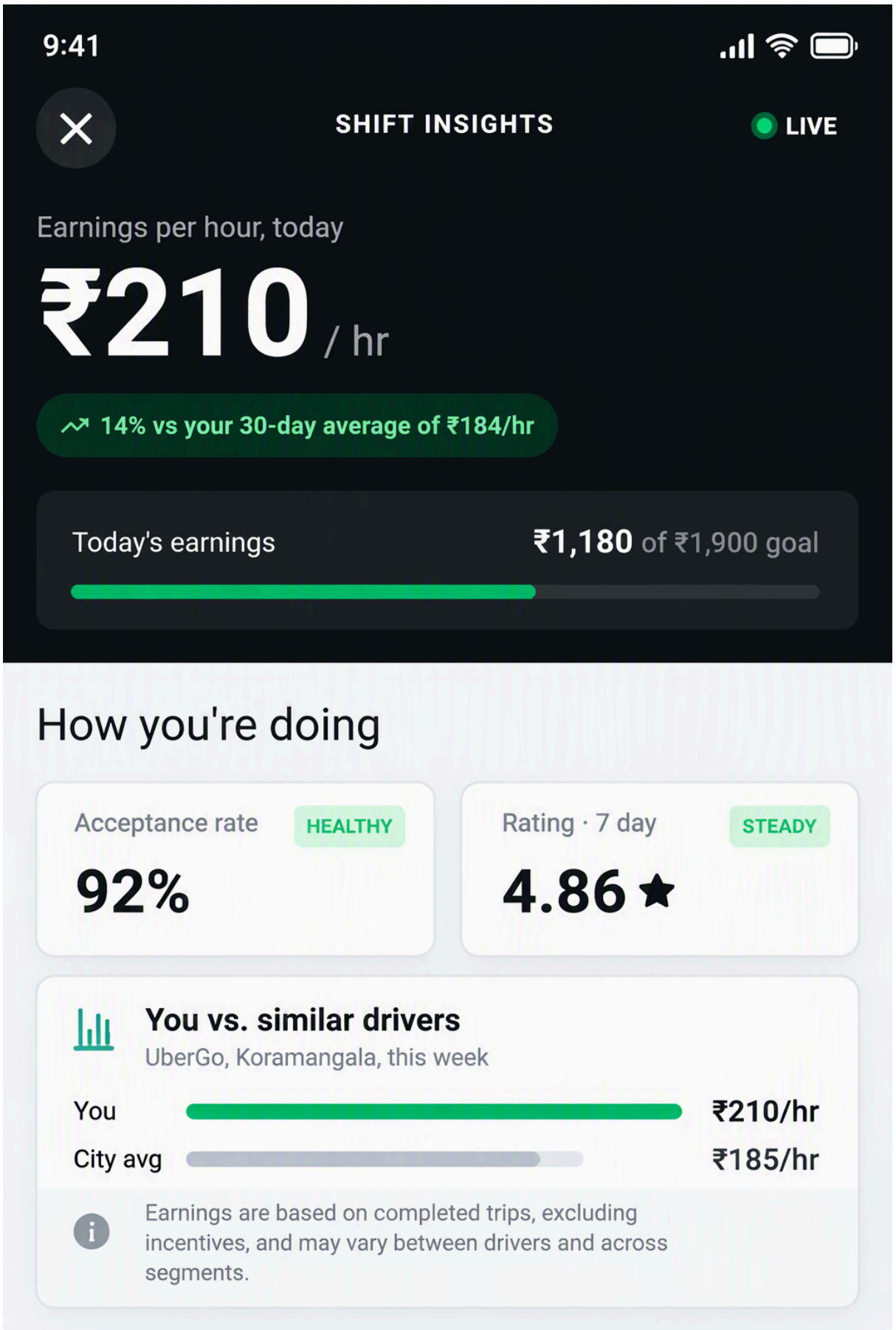
2.Shift Review Dashboard

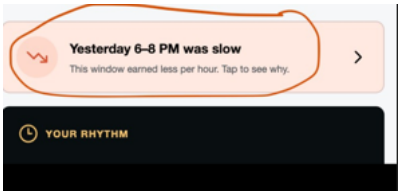
Shift Performance Analysis

The dashboard should help answer:

Was today's performance impacted by bad luck or by poor positioning?

This is one of the most important product capabilities.


9:41
SHIFT INSIGHTS
LIVE
Earnings per hour, today
₹210 / hr
14% vs your 30-day average of ₹184/hr
Today's earnings
₹1,180 of ₹1,900 goal
How you're doing
Acceptance rate
HEALTHY
92%
Rating · 7 day
STEADY
4.86
You vs. similar drivers
UberGo, Koramangala, this week
You
₹210/hr
City avg
₹185/hr
Earnings are based on completed trips, excluding incentives, and may vary between drivers and across segments.



Shift Diagnostics – Why Was My Shift Strong/Weak?

Drivers often struggle to understand whether a low-earning shift resulted from poor marketplace conditions ("bad luck") or suboptimal operational decisions such as driving in low-demand zones or during low-demand periods.

This section provides a root cause analysis of shift performance by breaking down the key factors that contributed positively or negatively to Earnings per Online Hour.

Key Questions Answered

- Why were my earnings above/below benchmark?
- Was demand unusually low today?
- Did I drive in the wrong zone?
- Was I active during the wrong time period?
- Did cancellations or idle time impact my earnings?
- What should I do differently next time?

Root Cause Analysis Framework

Factor	Driver Impact	Contribution
Zone Selection	Drove mostly in below-average demand zones	-12%
Time Selection	Missed evening peak demand period	-8%
Idle Time	Waited longer between trips than city average	-5%
Missed Surge Opportunity	Nearby surge zones not utilized	-4%
Acceptance Rate	Above average	+3%

Rating	Above average	+2%

Example

Today's Earnings: ₹185/hr

Comparable Driver Average: ₹235/hr

You earned 21% less than similar drivers.

Primary Reasons:

• Spent 65% of online time in low-demand zones (-12%)

• Missed evening peak demand window between 6 PM - 8 PM (-8%)

• Idle time was 18 min/hr vs city average of 9 min/hr (-5%)

Positive factors:

• Healthy acceptance rate (+3%)

• Strong driver rating (+2%)

UI\UX

Link - https://driver-next-level.lovable.app/

Dashboard Build

(Google AI Studio)-https://ai.studio/apps/3fa83bd4-2b34-486d-8a38-51f8415c6cad

Additional Notes

1.Net earnings would be a better success metric, but fuel consumption data is unavailable. As a proxy, we track deadhead distance (distance traveled without passengers).

2.Won't showing zone wise demand cause every data to rush to the same location?

This could be highly likely, to mitigate this, rather than showing all zones and locations only show the locations which closest to the driver (2-4km increase radius if demand is poor) because the driver would also consume and waste additional fuel for going to zones which are further away

3.What stops the demand alert from just pointing everyone at a shrinking opportunity? Pro-actively reduce the demand for that particular location/zone using a

predictive algorithm based on drivers accepting rides and moving towards those zones

4.Does the dashboard actively steer drivers toward under-served or remote areas?

No, the dashboard mainly provides signals to which locations have a higher per hour earning potential based on demand and price surges.

5.Could this dashboard push drivers into unhealthy hours?

A separate fatigue/performance metric (inversely proportional to the earnings per hour) to help the driver better understand if he is running tired and it would be better to rest and recover rather than accept rides.

6.How are "comparable drivers" defined for benchmarking?

The averages should be calculated within segments and not across segments, as well as segmentations need to be performed correctly. (A particular segment would depend on the city, ride vehicle(auto, ubergo, premier, xl, moto) – any other details/elements we can think of/add -> number of ride/working hours etc.)

6. What trade-offs does this design knowingly accept, rather than solve? Distance-based staggering reduces clustering but doesn't eliminate it — drivers roughly equidistant from a zone still get notified together. Drivers who happen to work near high-demand hubs get a natural, ongoing advantage simply from geography, not from any platform bias. These are named openly as accepted trade-offs, not gaps in the design